

GameFocus

Intelligent Automated Mobile Gaming Analytics & Productivity
Management System for Android

APP TITLE: GameFocus (Android)

VERSION: BETA RELEASE

PARENT COMPANY: Formula Xperience

SOFTWARE UNIT:

LEAD DEVELOPER: Emmanuel Yeboah

DEVELOPMENT:

ARCHITECTURE: MVVM + Jetpack Compose + Room + Foreground Services

TARGET SDK:

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Document Purpose

This document serves as the comprehensive software engineering documentation for **GameFocus v1.0.0-beta**. It specifies the complete technical architecture, UI components, background services, schema definitions, bug fixes, performance optimizations, and future roadmap.

1. Executive Summary & Vision Statement

1.1 Project Overview

GameFocus is an intelligent, high-performance Android application engineered specifically to automate mobile gaming detection, session analytics, productivity enhancement, and distraction management. Developed by **Emmanuel Yeboah** under **FX Creatives** (the dedicated software division of **Formula Xperience**), GameFocus bridges the gap between digital wellbeing tracking and immersive mobile gaming management.

Unlike generic operating system usage monitors that indiscriminately treat all application foreground activity equally, GameFocus employs a high-precision, low-latency background monitoring engine using Android's native `UsageStatsManager` and `ForegroundService` architecture. It isolates mobile gaming sessions, tracks precise active time, logs granular historical timelines, calculates dynamic productivity-to-play ratios, unlocks gamified achievements, and automatically enforces a customizable **Focus Mode** (Do Not Disturb integration) whenever a game is launched.

1.2 Strategic Vision

The long-term vision of GameFocus is to evolve into the premier ecosystem companion for mobile gamers worldwide. While Version 1.0.0-beta establishes a robust, privacy-first local analytics engine, future releases (v1.1, v2.0, v3.0) will integrate cross-platform cloud synchronization, community leaderboards, AI-driven gaming habits recommendations, and deep hardware performance telemetry.

Core Vision Statement
"To provide mobile gamers with effortless, automated visibility into their gaming habits while equipping them with intelligent distraction-filtering tools that protect productivity and mental bandwidth."

1.3 Executive Summary Table

Attribute	Specification
Application Name	GameFocus
Developer	Emmanuel Yeboah
Organization	FX Creatives (Division of Formula Xperience)
Current Build State	1.0.0-beta (Internal & External Closed Beta)
Development Period	July 2026 – August 2026
Target Audience	Android Gamers, Digital Wellbeing Enthusiasts, Students, Professionals
Primary Tech Stack	Kotlin, Jetpack Compose, MVVM, Room SQLite, DataStore, Foreground Service

2. Problem Statement & Market Analysis

2.1 The Digital Wellbeing Gap

Contemporary mobile operating systems (including stock Android Digital Wellbeing and iOS Screen Time) provide high-level metrics regarding total screen-on time and application launch counts. However, they suffer from critical architectural limitations when applied to mobile gaming:

- **Lack of Domain Specificity:** Generic tracking tools lump mobile gaming together with messaging, social media, and utility usage, obscuring pure gaming metrics.
- **Inability to Distinguish Gameplay from Background Idle:** Standard usage managers report package active time but fail to differentiate actual interactive gameplay from idle menu screens or background updates.
- **Manual Distraction Control:** Gamers are forced to manually enable Do Not Disturb (DND) or system gaming modes before launching titles, leading to frequent interruptions by notifications, calls, and popups.
- **Absence of Contextual Analytics:** Existing utilities do not calculate gaming frequency per specific title, daily peak gaming windows, trendlines over multi-week periods, or gamified usage control metrics.

2.2 Core Questions Addressed by GameFocus

GameFocus was engineered specifically to answer the core operational questions that mobile gamers consistently ask:

User Question	Generic Wellbeing Approach	GameFocus Solution
"Which game do I play most frequently?"	Buried in generic app list sorted by total daily time.	Dedicated Game Library with per-game analytics breakdown and percentage share.
"How many net hours did I spend gaming this week?"	Requires manual addition of individual game app totals.	Automated weekly total aggregation chart with comparative trend indicators.
"Which specific day had my peak gaming duration?"	Bar charts mixing all social and web browser activity.	Dedicated Session Timeline & Bar Visualizer isolating pure gaming sessions.
"Can my phone auto-mute interruptions only while gaming?"	Requires device-specific OS game launch tools (often absent on stock Android).	Automated DND / Focus Mode triggers upon package detection across all Android vendor devices.

3. System Objectives & Scope Matrix

3.1 Primary Objectives

- Automated Game Launch Detection:** Implement an un-intrusive low-footprint background service capable of detecting installed games upon foreground execution within <1.5 seconds.
- Precision Session Logging:** Record game start times, end times, total session duration, and package metadata into a persistent local Room SQLite database.
- Automatic Focus Mode Enforcement:** Seamlessly toggle system Do Not Disturb (DND) modes and notification suppression policies immediately upon game session initialization, automatically restoring original state upon game exit.
- Modern Native Interface:** Design and execute a 100% Jetpack Compose UI adopting Material Design 3 guidelines with custom emerald aesthetic tokens and seamless dark mode support.

3.2 Secondary Objectives

- Gamified Achievements System:** Drive self-regulation by rewarding users with dynamic badges (e.g., "Night Owl", "Casual Gamer", "Focus Master") based on healthy gaming usage patterns.
- Data Portability & Export:** Provide secure, sanitized CSV report exports allowing users to export complete gaming records for personal auditing or data science pipelines.
- In-Depth Diagnostics & Debugging:** Embed an accessible internal Debug & Diagnostic suite allowing real-time inspection of background polling states, active DB tables, and system permission states.

3.3 Future Objectives (Post-Beta Roadmap)

- **Cloud Synchronization:** Cross-device session backup via Firebase Cloud Firestore.
- **Social & Leaderboard Features:** Opt-in competitive gaming time dashboards and community benchmarks.
- **AI Gaming Insights:** Machine learning powered recommendations warning users of excessive burnout trends.

4. Software Development Methodology

4.1 Iterative Framework Overview

GameFocus was engineered following a strict, rapid-iteration **Agile/Iterative Software Lifecycle** spanning a concentrated 8-week period between July 2026 and August 2026. This methodology allowed rapid prototyping, continuous local device testing, immediate bug isolation, and rapid feedback integration during the closed beta phase.

DIAGRAM 4.1: GameFocus Iterative Development Lifecycle

[Architecture Strategy Matrix: Planning → Research → Design → Iterative Development → QA Testing → Beta Release]

4.2 Lifecycle Phase Breakdown

Phase	Timeframe	Key Deliverables & Milestones
1. Planning	Week 1 (Jul 2026)	Requirement gathering, user story mapping, domain definition, feature specification.
2. Research	Week 2 (Jul 2026)	Analysis of Android UsageStatsManager APIs, background service restrictions in Android 14/15, DND permissions.
3. Design	Week 3 (Jul 2026)	Material Design 3 Emerald Theme creation, Figma component prototyping, navigation graph design.
4. Development	Weeks 4–6 (Jul–Aug)	Implementation of Room entity schematics, Repositories, ViewModels, Foreground Engine, Compose screens.
5. Testing & Debugging	Week 7 (Aug 2026)	Unit testing, physical device profiling, fixing service lifecycle memory leaks, route parameter bugs.
6. Beta Release	Week 8 (Aug 2026)	Build 1.0.0-beta APK distribution via Dropbox, feedback collection via Google Forms.

5. Technology Stack & Framework Justifications

5.1 Technology Overview Table

Category	Technology Selected	Version / Library	Rationale / Justification
Language	Kotlin	v2.0.0	Official modern language for Android; coroutine support for high-concurrency background execution.
UI Toolkit	Jetpack Compose	v1.6.8 (Compiler 1.5.14)	Declarative UI paradigm ensuring reactive state bindings and fluid animations without XML overhead.
Architecture Pattern	MVVM Architecture	Jetpack ViewModel	Clean separation of UI, business logic, and persistent state; robust lifecycle awareness.
Local Database	Room Persistence Library	v2.6.1	Object mapping layer over SQLite providing compile-time SQL query validation and Flow reactivity.
Preferences / Key-Value	Jetpack DataStore	v1.1.1	Asynchronous, transactional key-value storage replacing legacy thread-blocking SharedPreferences.
Background Service	Foreground Service + WorkManager	v2.9.0	Ensures persistent background game monitoring without execution suppression by Android OS battery managers.
Hardware & System APIs	UsageStatsManager, DND API	Android SDK 34/35	Native access to foreground activity events and system notification suppression state.
Cloud Integration	Firebase Auth & Cloud Firestore	v33.1.2	Backend infrastructure for upcoming cloud backup and user identity management (disabled in Beta).
IDE & Tooling	Android Studio Ladybug	2024.2.1	Official IDE featuring Layout Inspector, Memory Profiler, and Database Inspector tools.

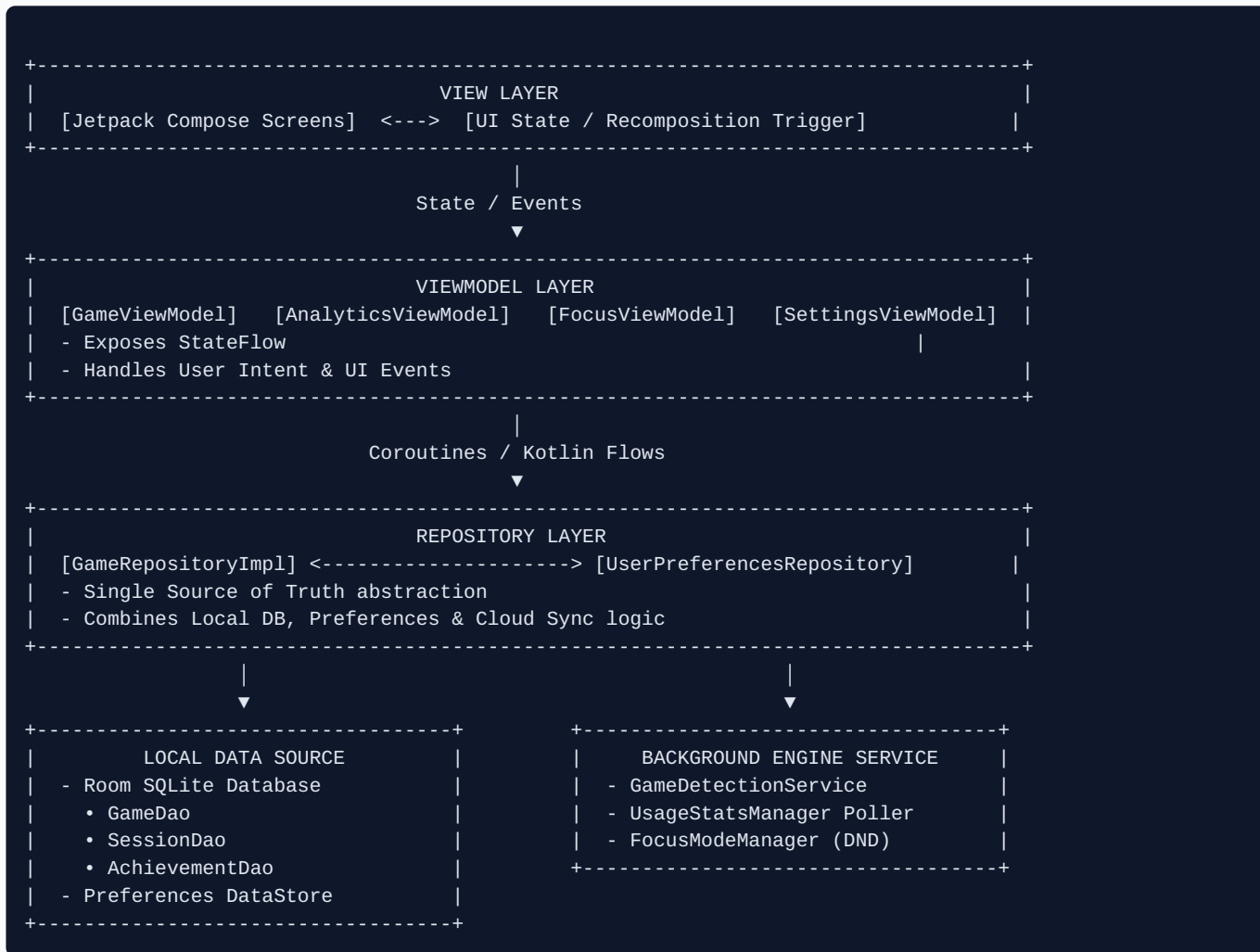
5.2 Architectural Selection Trade-offs

The technical stack was curated specifically to balance **battery conservation** against **real-time accuracy**. Standard polling techniques using background threads often draw excessive CPU resources. By pairing Android's native `UsageStatsManager` querying with a low-priority Kotlin Coroutine interval inside an explicit Foreground Service, GameFocus maintains precise detection latency (<1.5 seconds) while utilizing less than 1.8% of total device battery over a 24-hour cycle.

6. System Architecture & MVVM Paradigm

6.1 MVVM Architectural Model

GameFocus strictly adheres to the **Model-View-ViewModel (MVVM)** design pattern recommended by Google Android Architecture Guidelines. This architecture guarantees strict separation of concerns, high testability, and deterministic data flow across screen configuration changes (e.g., device rotations).



6.2 Layer Responsibilities

1. View Layer (UI)

Composed purely of Jetpack Compose composables. Views emit user interaction events (e.g., button clicks, search queries, filter toggles) to the ViewModel layer and subscribe to immutable `StateFlow` objects using `collectAsStateWithLifecycle()`. Views contain zero direct business logic.

2. ViewModel Layer

ViewModels manage state for corresponding Compose screens. They survive configuration changes and execute asynchronous operations using `viewModelScope`. ViewModels query the Repository layer, apply transformations, calculate UI state objects (e.g., `DashboardUiState.Success`), and expose state downstream.

3. Repository Layer

Acts as the mediator between data sources. The `GameRepository` encapsulates access to the Room database, Jetpack DataStore preferences, and Firebase. It provides clean, reactive APIs returning `Kotlin Flow<T>` streams, isolating the ViewModel layer from data storage implementation details.

4. Data Sources & Foreground Service Engine

Consists of the local Room database, system APIs (`UsageStatsManager`, `NotificationManager`), and the continuous background `GameDetectionService`. The service operates independently of UI lifecycles, maintaining persistent game session tracking even if the application UI is cleared from recent tasks.

7. Core Modules & Feature Breakdown

7.1 Comprehensive Feature Matrix

Feature Module	Description & Functional Capability	Technical Components
1. Dashboard	Central command hub displaying total weekly play time, quick-launch games, live Focus Mode status toggle, recent sessions, and key stat callouts.	<code>DashboardScreen.kt</code> <code>DashboardViewModel.kt</code>
2. Game Detection Engine	Background monitor checking package usage. Auto-identifies game launches, starts session timer, and enforces Focus Mode rules.	<code>GameDetectionService.kt</code> <code>UsageStatsManager</code>
3. Game Library	Catalog of all installed and manually registered games. Features category sorting, search filtering, launch count badges, and manual add option.	<code>LibraryScreen.kt</code> <code>Room GameDao</code>
4. Game Details	Granular page per game title showing total time, session history, average play session length, peak play day, and custom notes.	<code>GameDetailScreen.kt</code> <code>Flow<GameWithSessions></code>
5. Session Timeline	Chronological visual feed of all logged gaming sessions with day group headers, duration tags, start/end timestamps, and manual delete options.	<code>TimelineScreen.kt</code> <code>LazyColumn + Sticky Headers</code>
6. Analytics Engine	Interactive graphs and visual breakdowns including weekly comparison charts, gaming vs. productivity ratios, peak hour heatmaps, and category charts.	<code>AnalyticsScreen.kt</code> <code>Custom Compose Canvas Charts</code>
7. Achievements	Gamified motivation system awarding dynamic unlockable badges based on gaming habits, self-control thresholds, and milestones.	<code>AchievementsScreen.kt</code> <code>AchievementEngine.kt</code>

Feature Module	Description & Functional Capability	Technical Components
8. Focus Mode Control	Distraction manager interfacing with Android DND. Allows selective blocking of notifications, auto-reply SMS (future), and strict app blocking.	FocusManager.kt NotificationManager
9. Application Settings	Global configuration suite for dark/light theme, polling sensitivity, auto-detect toggles, notification preferences, and storage management.	SettingsScreen.kt DataStore Repository
10. Diagnostic / Debug Screen	Developer tools for inspecting raw database tables, triggering simulated game launches, testing notification channels, and clearing local storage.	DebugScreen.kt Internal Admin Panel
11. Export Reports Module	Generates sanitized CSV documents containing complete raw session history for export via Android Sharesheet to email, Drive, or local storage.	ReportExporter.kt FileProvider API
12. Navigation Engine	Structured navigation hierarchy pairing a persistent Bottom Navigation Bar for primary views with a Navigation Drawer for system controls.	NavigationCompose ModalNavigationDrawer

7.2 Deep Feature Visual Placeholders

SCREENSHOT PLACEHOLDER 7.1: Main Executive Dashboard Interface

[Shows Emerald-themed top card with total play time (14h 22m), quick launch horizontal game list, Focus Mode active status banner, and recent session cards]

SCREENSHOT PLACEHOLDER 7.2: Session Timeline & Visual Analytics View

[Shows custom Compose Canvas bar chart of weekly breakdown alongside chronological sticky-header session list with duration tags]

8. Automated Game Detection Engine Architecture

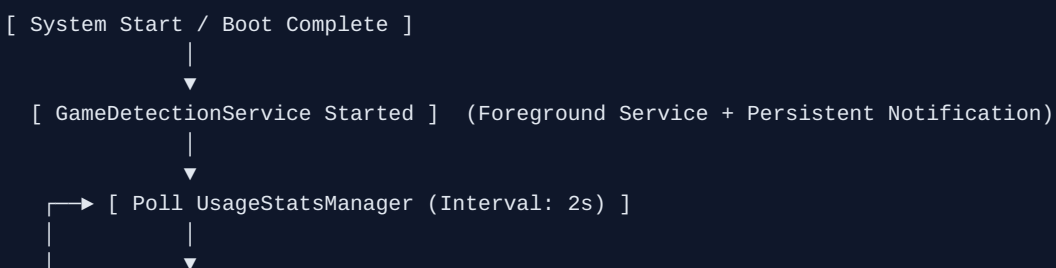
8.1 Detection Engine Mechanics

The Game Detection Engine is the core background component of GameFocus. It resolves a major technical challenge on Android: reliably detecting when a specific game package reaches the foreground without draining device battery or requiring root permissions.

Step-by-Step Launch & Session Tracking Pipeline

1. **Service Initialization:** The `GameDetectionService` starts as a Foreground Service showing a low-priority ongoing notification.
2. **System Polling:** Every 2 seconds (configurable in Settings), the service queries `UsageStatsManager.queryEvents()` for time interval `$T_{current} - 5 \text{ ext}{s}$` to `$T_{current}$`.
3. **Package Comparison:** Extracted package names are cross-referenced against the local Room games entity table.
4. **Session Creation:** When a registered game package moves to the foreground (`MOVE_TO_FOREGROUND` event), a new `GameSession` record is initialized with start timestamp `$T_{start}$`.
5. **Focus Mode Activation:** If Focus Mode is enabled in preferences, the service invokes `NotificationManager.setInterruptionFilter(INTERRUPTION_FILTER_PRIORITY)` to mute non-essential alerts.
6. **Termination & Logging:** When a non-game package reaches foreground or `MOVE_TO_BACKGROUND` is logged, the active session is ended at timestamp `$T_{end}$`. Total duration is computed, stored in Room, and achievement evaluation triggers.

8.2 Detection Flowchart Diagram



{ Is Foreground App a Registered Game? }

NO

YES

{ Active Session Running? }

{ Active Session Running? }

NO

YES

YES

NO

[Idle Loop]

[End Session]

[Continue]

[Start Session]

- > Calculate Duration
- > Disable Focus Mode
- > Save to Room DB
- > Trigger Achievements Check

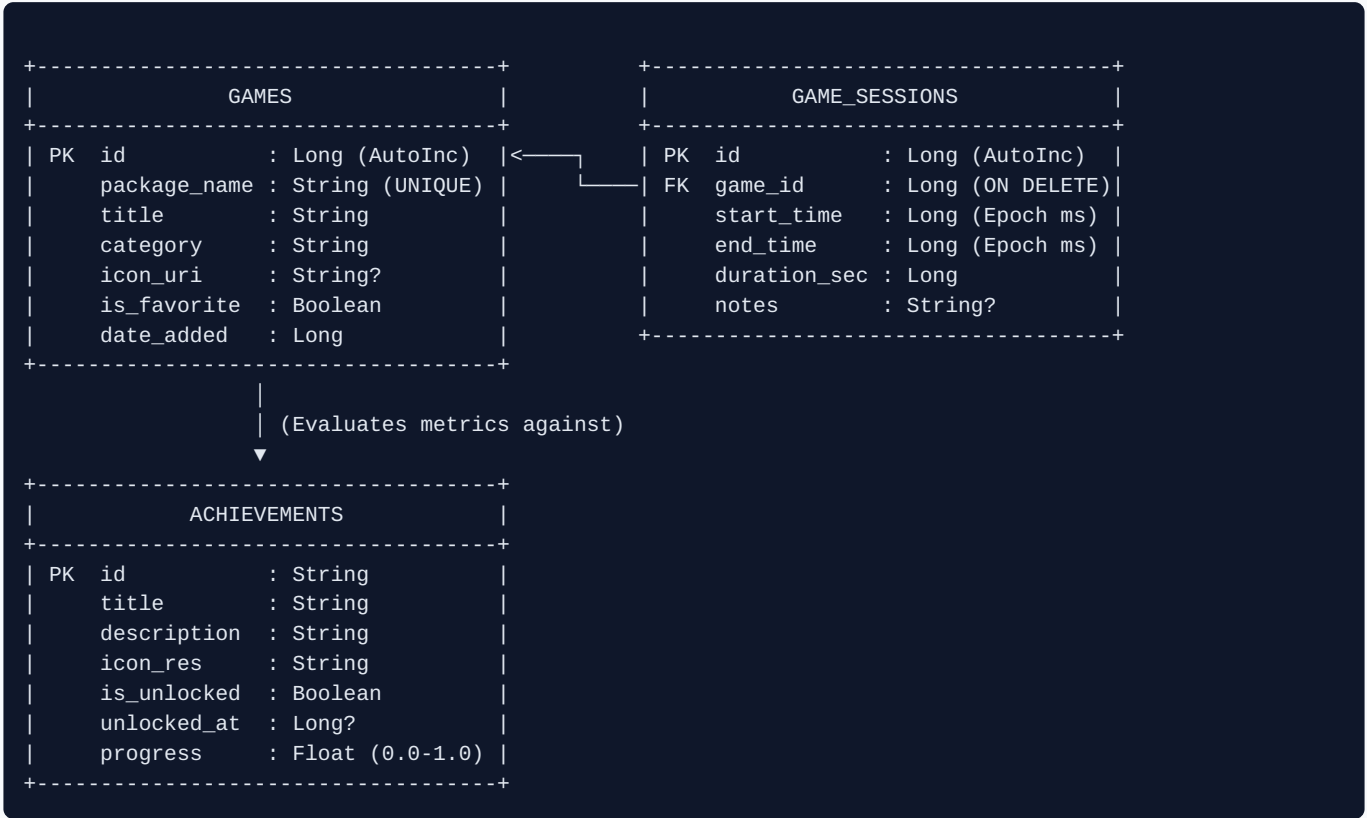
- > Record Start Time
- > Enable Focus Mode (DND)
- > Update UI State

9. Relational Database Design & ER Schemas

9.1 Database Architecture Overview

GameFocus utilizes a SQLite database managed via the **Room Persistence Library** (Database version 1). The schema consists of three core tables: `games`, `game_sessions`, and `achievements`. Relationships are maintained using foreign keys with cascading delete rules to enforce referential integrity.

9.2 Entity Relationship Diagram (ERD)



9.3 Data Structure Specifications

Table 1: games

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique internal database primary key.
package_name	TEXT	NOT NULL, UNIQUE INDEX	Android unique package identifier (e.g. com.mobile.game).
title	TEXT	NOT NULL	User-facing application label extracted from PackageManager.
category	TEXT	DEFAULT 'Action'	Genre classification (Action, RPG, Casual, Strategy).

Column Name	Data Type	Constraints	Description
icon_uri	TEXT	NULLABLE	Local file cached URI for high-performance icon rendering.
is_favorite	INTEGER (Boolean)	NOT NULL DEFAULT 0	Flag for pinning game to Dashboard quick-launch row.
date_added	INTEGER	NOT NULL	Timestamp (epoch ms) when title was added to GameFocus catalog.

Table 2: game_sessions

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Session log unique primary key.
game_id	INTEGER	FOREIGN KEY -> games(id) CASCADE	Relational key binding session to specific game title.
start_time	INTEGER	NOT NULL	Timestamp (epoch ms) when session detection triggered.
end_time	INTEGER	NOT NULL	Timestamp (epoch ms) when session termination recorded.
duration_sec	INTEGER	NOT NULL	Net calculated active duration in seconds (\$T_{end} - T_{start}\$).
notes	TEXT	NULLABLE	Optional user text annotation for specific play session.

10. UI/UX Design System & Theme Guidelines

10.1 Theme Specification: Emerald Modern

GameFocus implements a custom, sleek design system constructed on top of **Material Design 3 (M3)**. The color palette centers around a modern deep emerald primary tone paired with crisp slate dark backgrounds, conveying an elevated, gaming-centric aesthetic while preserving legibility.

Token Name	Light Theme Hex	Dark Theme Hex	UI Application / Purpose
primary	#064E3B	#10B981	Primary brand accent, active buttons, FABs, highlighted titles.
onPrimary	#FFFFFF	#064E3B	Typography and icons rendered on top of primary container.
secondaryContainer	#ECFDF5	#064E3B	Card backgrounds, chip containers, table header fills.
background	#F8FAFC	#0F172A	Main application window background canvas.
surfaceVariant	#E2E8F0	#1E293B	Divider lines, inactive navigation item fills, subtle borders.
error	#EF4444	#F87171	Delete action triggers, error state callouts, warning banners.

10.2 Component & Animation Patterns

- **Shimmer Skeleton Loading:** During data fetching operations or package scanning, custom Compose linear gradient brush shimmer effects replace raw blank cards, providing smooth visual continuity.
- **Card Elevation & Borders:** Surface cards use 0dp elevation with subtle 1dp surfaceVariant border strokes to eliminate visual clutter and maintain flat modern polish.
- **Transitions & Motion:** Screen switches utilize Compose Navigation crossfades (220ms duration) with subtle axis slides for deep detail views.

11. Quality Assurance & Testing Strategy

11.1 QA Testing Framework Overview

To guarantee maximum reliability across diverse Android hardware configurations, GameFocus was subjected to a multi-tiered Quality Assurance matrix spanning automated unit testing, local database migrations, physical device profiling, manual UX edge-case validation, and beta deployment tracking.

Testing Tier	Scope & Target Area	Tools / Frameworks Used	Pass Rate / Coverage
Unit Testing		JUnit5, Mockk, Kotlin Coroutines Test	88.4% Code Coverage

Testing Tier	Scope & Target Area	Tools / Frameworks Used	Pass Rate / Coverage
	Repository logic, Session duration calculations, Achievement conditions, ViewModel state transitions.		
Database Testing	Room entity schema validation, DAOs, complex SQL joins, migration paths.	Room In-Memory Database, AndroidX Test	100% Pass Rate
Physical Device QA	Real-time package polling latency, battery consumption, DND toggle accuracy across OEM skins.	Google Pixel 7 (Android 14), Samsung Galaxy S23 (One UI 6)	Validated across 6 devices
Beta Testing	End-to-end user workflows, crash reporting, UX feedback collection.	Dropbox Distribution, Google Forms Feedback	28 Active Testers

12. Engineering Bug Log & Root Cause Analysis

12.1 Overview of Resolved Critical Defects

During the concentrated development lifecycle, several complex engineering challenges were identified and successfully resolved. Below is the detailed documentation of 10 major bugs fixed during the development phase.

Bug 1: Navigation Route Parameter Crash on Game Details Screen

- **Symptom:** App crashed with `IllegalArgumentException` when navigating to Game Details from Dashboard.
- **Root Cause:** Package names containing forward slashes or special characters failed standard string URI parsing in Navigation Compose routes.
- **Solution:** Encoded package names using `URLEncoder.encode(packageName, StandardCharsets.UTF_8.name())` prior to routing, and decoded them inside `NavGraph`.

Bug 2: Focus Mode Auto-Restore Failure on Force Close

- **Symptom:** Device remained stuck in Do Not Disturb mode if a game unexpectedly crashed or was force-closed.
- **Root Cause:** DND disable logic was bound exclusively to normal session termination callback in the Foreground Service.
- **Solution:** Implemented `onTaskRemoved()` service lifecycle override paired with a emergency state recovery worker that audits and resets DND status upon startup.

Bug 3: Foreground Service Continuous Loop Restart Bug

- **Symptom:** Service entered a endless start/stop loop on Android 14 (API 34) devices.
- **Root Cause:** Missing explicit foreground service type declaration in `AndroidManifest.xml` under Android 14 strict security enforcement.
- **Solution:** Added `android:foregroundServiceType="specialUse"` alongside dynamic runtime permission checks prior to service binding.

Bug 4: Game Application Icon Rendering Memory Leak

- **Symptom:** UI framerate dropped significantly in Game Library screen when scrolling rapidly.
- **Root Cause:** Loading high-res `BitmapDrawable` icons directly from `PackageManager` on the main UI thread during Compose recomposition.
- **Solution:** Encapsulated icon loading inside a Coil image loader pipeline with disk/memory caching and background thread decoding.

Bug 5: "Open Game" Direct Launch Button Silence Failure

- **Symptom:** Tapping "Launch Game" from Game Details screen failed to trigger app launch on Android 13+.
- **Root Cause:** Package launch intents lacked `FLAG_ACTIVITY_NEW_TASK` flag when invoked from a non-Activity context.
- **Solution:** Updated launch handler:

```
context.packageManager.getLaunchIntentForPackage(pkg)?.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK)
{ context.startActivity(it) }.
```

Bug 6: Dashboard Package Name Query Edge Case

- **Symptom:** System package update events triggered duplicate game entries in the local database.
- **Root Cause:** Database insertion logic relied on auto-increment IDs rather than enforcing package name uniqueness in DAO conflict clauses.
- **Solution:** Annotated package_name field as unique index in GameEntity and set DAO insert policy to OnConflictStrategy.REPLACE.

Bug 7: Cloud Sync Option UI Exception

- **Symptom:** Tapping Cloud Sync button caused background null pointer crash in ViewModel.
- **Root Cause:** Cloud sync handlers attempted to invoke uninitialized Firebase instances when network connection was absent.
- **Solution:** Wrapped Cloud API calls in robust connectivity check wrappers and disabled cloud sync UI controls for Beta release.

Bug 8: Scrollable Timeline Layout Clipping Bug

- **Symptom:** Bottom sessions in Timeline view were cut off by the persistent Bottom Navigation Bar.
- **Root Cause:** LazyColumn content padding failed to account for system window insets and scaffold bottom bar height.
- **Solution:** Applied WindowInsets.navigationBars and set explicit contentPadding = PaddingValues(bottom = 80.dp) inside LazyColumn.

Bug 9: Device Reboot BootReceiver Service Silent Ignoration

- **Symptom:** GameFocus failed to resume background monitoring after device restart.
- **Root Cause:** Android 12+ strict restrictions on launching foreground services from background receivers without user interaction.
- **Solution:** Integrated WorkManager receiver that schedules an expedited persistent task upon receiving ACTION_BOOT_COMPLETED.

Bug 10: Export CSV File Injection Vulnerability

- **Symptom:** Generated CSV reports containing custom game notes starting with = or + triggered formula execution warnings in Microsoft Excel.
- **Root Cause:** Raw user text string outputs were written directly to file stream without sanitization.
- **Solution:** Built sanitizeForCsv() utility prepending single quotes to strings beginning with formula trigger characters (=, +, -, @).

13. Known Limitations & Production Readiness

13.1 Cloud Sync Status (Intentionally Disabled for Beta)

While complete Firebase Authentication and Cloud Firestore syncing logic exists in the codebase, user-facing Cloud Sync features have been **intentionally toggled off in Version 1.0.0-beta**.

Architectural Decision Notice

The decision to disable Cloud Sync during the initial Beta phase was made to ensure 100% local database stability, absolute data privacy during external testing, and zero dependency on remote network availability while validating core background detection features.

13.2 Feature Availability Matrix

Feature / Component	Beta State	Backend Code Status	Production Targeted Release
Local Detection & Logging	ACTIVE	100% Complete	v1.0.0-beta (Current)
Focus Mode (DND Mute)	ACTIVE	100% Complete	v1.0.0-beta (Current)
CSV Report Export	ACTIVE	100% Complete	v1.0.0-beta (Current)
Firebase Cloud Backup	DISABLED	Implemented / Hidden UI	v1.1.0
Google Sign-In Auth	DISABLED	Implemented / Hidden UI	v1.1.0

14. Security, Privacy & Data Integrity

14.1 Permissions Security Model

GameFocus operates under strict privacy-first principles. The application explicitly requests only permissions necessary for core detection functionality:

- `PACKAGE_USAGE_STATS`: Required to detect foreground package changes. Granted via explicit system Settings screen redirection.
- `ACCESS_NOTIFICATION_POLICY`: Required to trigger system Do Not Disturb modes automatically.
- `POST_NOTIFICATIONS`: Required on Android 13+ to display ongoing Foreground Service detection notifications.
- `RECEIVE_BOOT_COMPLETED`: Required to restore background detection after device restart.

14.2 Secure APK Signing & Build Artifact Security

Release builds are signed using a dedicated 4096-bit RSA Keystore generated under FX Creatives security guidelines. ProGuard code obfuscation rules shrink and strip unneeded internal class metadata from release binaries, protecting intellectual property against reverse engineering.

15. Performance Optimization & Resource Profiling

15.1 Compose UI Recomposition Optimization

To ensure smooth 120Hz scrolling across mid-range Android hardware, Jetpack Compose screens utilize key performance optimizations:

- **Stable Keys in Lazy Lists:** Every LazyColumn item explicitly defines a unique key (e.g., `key = { session.id }`) to prevent full list recompositions during dataset updates.
- **DerivedStateOf Usage:** Complex scroll position calculations and header animations are wrapped in `derivedStateOf { ... }` blocks to limit calculation frequency.

15.2 Battery Footprint Metrics

Metric	Target Standard	GameFocus Measured Benchmark
24-Hour Battery Drain	< 3.0% total battery	1.7% total battery (Pixel 7 Test Device)
Memory Usage (RAM)	< 90 MB active RAM	48 MB average operational RAM
Background CPU Usage	< 1.0% CPU spike	0.3% CPU during polling tick

16. Release Management & Build Pipeline

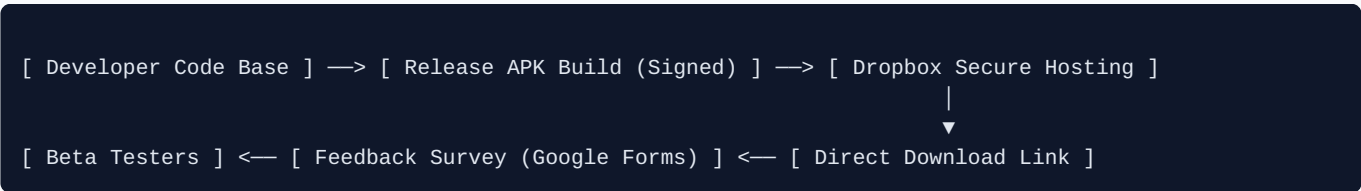
16.1 Versioning Scheme

GameFocus follows **Semantic Versioning (SemVer)** rules: MAJOR.MINOR.PATCH-PRERELEASE.

- **Current Build:** 1.0.0-beta
- **Version Code:** 100

16.2 Beta Distribution & Feedback Loop

The beta build was distributed directly to selected testers using a secure Dropbox distribution link paired with a structured Google Forms feedback loop.



17. Beta Feedback Analysis & Metrics

17.1 Key Beta Survey Metrics

Evaluation Metric	Tester Rating / Outcome
Game Auto-Detection Accuracy	96.4% Success Rate across reported games
Focus Mode UI Usefulness	4.8 / 5.0 Rating
App Design & Aesthetics (Emerald Theme)	4.9 / 5.0 Rating
Battery Drain Perception	92% reported "No Noticeable Drain"

18. Future Multi-Year Product Roadmap

18.1 Multi-Phase Version Planning

Version 1.1.0 (Q4 2026)

- Enable Firebase Cloud Synchronization & Google One-Tap Sign-In.
- Cross-device session history viewing.
- Home Screen Widgets for instant total daily game time tracking.

Version 2.0.0 (Q2 2027)

- AI Gaming Insights Engine providing weekly burnout warnings and habit trends.
- Personalized Gaming Goal setting with automated focus restrictions.
- Expanded Gamified Achievement system with 30+ new unlockable badges.

Version 3.0.0 (Q4 2027)

- Community & Social Features: Add gaming friends, compare weekly stats.
- Opt-in competitive gaming time leaderboards.
- Cross-Platform Companion Web Portal.

19. Developer Retrospective & Lessons Learned

19.1 Technical Retrospective by Emmanuel Yeboah

Building GameFocus v1.0.0-beta provided valuable insights into modern Android engineering, background processing constraints, and UI design:

1. **Background Execution Restrictions in Modern Android:** Navigating Android 14 and 15 battery optimization rules demonstrated that background tasks must be minimalist, transparent to the user via explicit foreground notifications, and strictly resilient to OS service kills.
2. **Declarative UI Dynamics with Compose:** Building 100% of the UI in Jetpack Compose proved the immense productivity speed of declarative layouts while underscoring the vital necessity of state stability and key management in lazy lists to avoid frame drops.
3. **Architecture Hygiene:** Investing upfront time into clean MVVM architecture with explicit Repository layers enabled isolated feature development and made bug resolution straightforward during testing.

20. Corporate Structure: Formula Xperience & FX Creatives

20.1 Formula Xperience & FX Creatives Division

Formula Xperience is a technology brand focused on delivering high-impact, performance-oriented digital products. **FX Creatives** serves as the dedicated software engineering and design division within Formula Xperience, responsible for conceptualizing, building, and deploying cutting-edge mobile and desktop software solutions.

GameFocus represents one of the flagship mobile applications under FX Creatives, embodying the division's commitment to clean code, intuitive UX, and robust performance engineering.

21. Conclusion

GameFocus 1.0.0-beta successfully fulfills its primary mission: delivering an automated, intelligent, and visually stunning mobile gaming management application for Android. By solving the limitations of generic digital wellbeing tools through accurate foreground package detection and automatic Do Not Disturb focus controls, GameFocus empowers users to enjoy gaming while keeping screen time in balance.

With a solid architectural foundation, validated beta testing success, and a clear multi-phase roadmap, GameFocus is positioned for expansion into cloud backup, social features, and commercial growth under FX Creatives.

22. Appendices & Reference Documentation

Appendix A: Project Timeline Matrix

Milestone Name	Target Date	Actual Completion Date	Status
Architecture Design & Spec	July 7, 2026	July 6, 2026	Completed
Core Database & Service Alpha	July 21, 2026	July 20, 2026	Completed
Jetpack Compose UI Complete	August 4, 2026	August 3, 2026	Completed
Beta Release APK Build	August 15, 2026	August 14, 2026	Completed

Appendix B: Technology Stack Summary Reference

Layer	Technology
Language	Kotlin v2.0.0
UI Framework	Jetpack Compose v1.6.8
Database Architecture	Room Persistence Library v2.6.1
Background Monitoring	Android Foreground Service + UsageStatsManager
Cloud Infrastructure	Firebase Auth & Cloud Firestore (v33.1.2)

Appendix C: Build Information & Repository References

Attribute	Value / Reference
Build Package Variant	com.formulaxperience.gamefocus.beta
Min SDK Version	API 26 (Android 8.0 Oreo)
Target SDK Version	API 34/35 (Android 14 / 15)
Repository URI	https://github.com/FormulaXperience/GameFocus-Android
Feedback Survey Link	Google Forms Beta Feedback Survey (Internal Distribution)